easant
now that
t in $\mathbb{Q}_7$ is

$, \frac{4}{1}, \frac{5}{1}, \frac{6}{1}$

$5, 6.$

r arithmetic

$= \mathbb{Z}/7\mathbb{Z}$

Exercise: If you do same for 6 instead of 7, $\mathbb{Q}_6 = \{0,1,2,3,4,5\} \pmod 6$ is a ring, but not a field.

This ring has zero divisors

$$2 \times 3 = 6 = 0$$

Another important field:

$\mathbb{Q}(\sqrt{2})$: $a + b\sqrt{2}$ $\qquad (\sqrt{2})^2 = 2$

$$(2+5\sqrt{2})(3-\sqrt{2}) = -4 + 13\sqrt{2}.$$